

Introduction

Microplastics are currently ranked the second most emerging environmental issue behind global warming, due to a large increase in plastic usage in the past several decades. Plastics degrade into small microplastic particles, but do not decompose. Plastic additive chemicals leach out of microplastics within the environment, causing irreversible damage to soil, an important natural resource. Microplastics cause oxidative stress on plants, leading to decreased levels of plant growth development. Plants produce antioxidants to increase tolerance and neutralize oxidative stress and its effects. Drench solution application of antioxidants on plants can enhance tolerance to stressors. However, this has not been tested on *Brassica rapa*.



Photo taken by finalists

Fig 1. All Flats of *Brassica rapa*

Purpose

To determine if L-ascorbic acid, an antioxidant, mitigates the negative effects of polystyrene microplastics (PS MPs/styrofoam) and to identify the most effective concentration. With the large increase in microplastic pollution in soil, crops and produce are being contaminated and negatively impacted, resulting in human ingestion of microplastic particles. The real world value of this experiment is that it could establish a natural way to protect crops from damage, leading to less food scarcity.

Use of L- Ascorbic Acid to Mitigate Effect of Polystyrene Microplastics on Growth of *Brassica rapa*

Methodology

Groups:
-4 experimental - Negative, Positive: 1, 2, 3
-1 control

150 pots total and 15 pots per flat

300mg of styrofoam was mixed into soilless media of each experimental pot. 3 *Brassica rapa* seeds were sown into each pot. The flats were placed into a growth chamber at 27.0°C for three days. For the first concentration, 50mg of L-ascorbic acid and 1L of reverse osmosis water was measured. This was repeated twice using 100mg and 150mg of L-ascorbic acid respectively.

Drench Solution of L-ascorbic acid Every 2 Days:

- Control: 0%
- Negative: 0%
- Positive 1: 1 mL 0.0015% L-ascorbic acid
- Positive 2: 1 mL 0.01% L-ascorbic acid
- Positive 3: 1mL 0.015% L-ascorbic acid

After three weeks, each plant was cut at the soil line to measure biomass in grams.



Photo taken by finalists

Fig 2. Applying Treatment



Photo taken by finalists

Fig 3. Measuring Biomass of *Brassica rapa*



Photo taken by finalists

Fig 6. *Brassica rapa* Positive 2 Pots

Discussion

A parametric ANOVA test was conducted using the data collected, giving a p-value of 0.02, establishing statistically significant results. The results suggest that the median biomass of the *Brassica rapa* increases as the concentration of L-ascorbic acid increases. Thus, the highest concentration of 0.015% was the most effective at mitigating the negative effects of PS MPs.

The graph comparing the Negative group with Positive 3 shows the best correlation out of all positive groups. Since the line of best fit is downwards, the correlation is gradually decreasing. The p-value is 0.11, indicating that there is some relationship between both variables.

Results

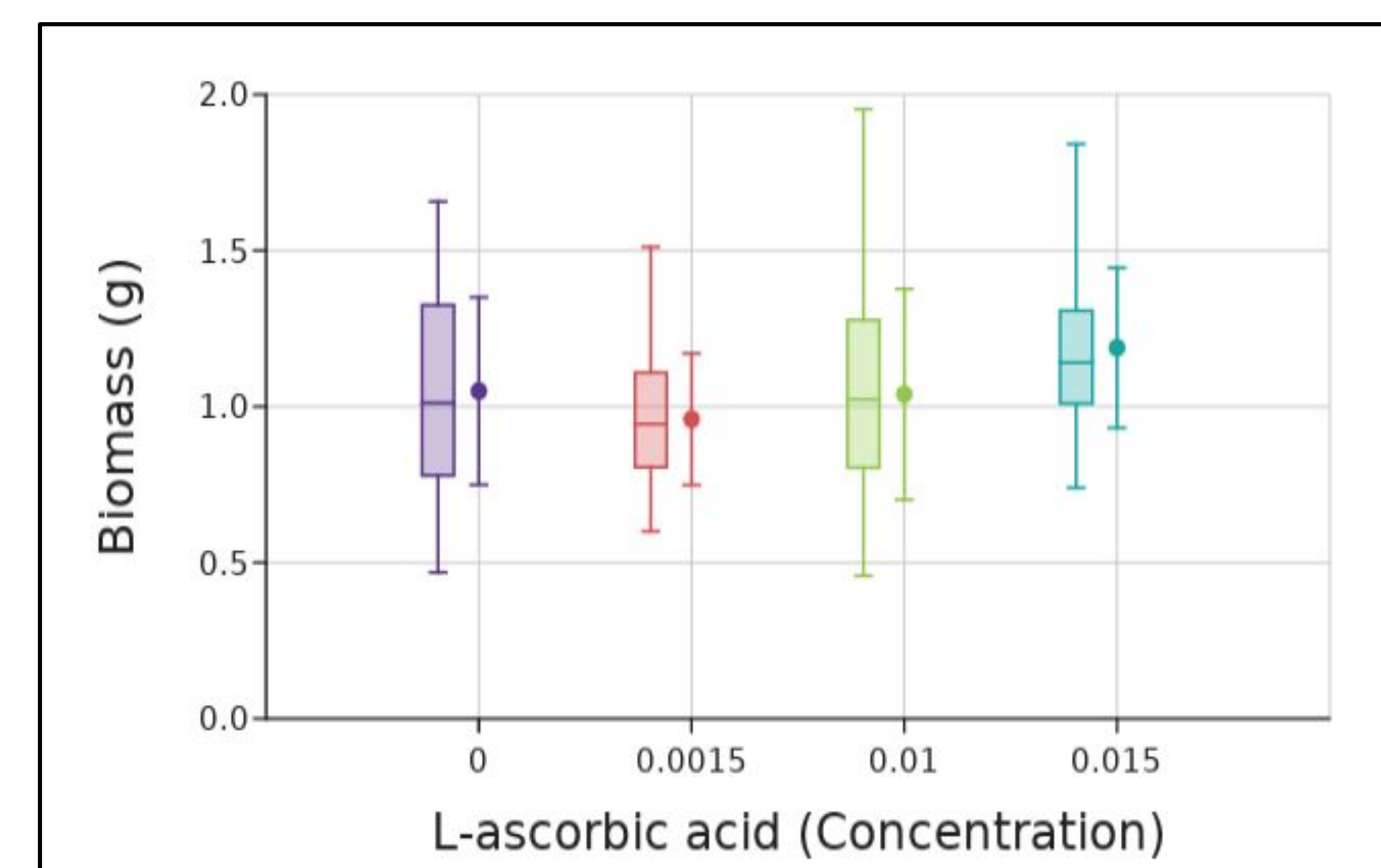


Chart created by finalists using DataClassroom

Fig 4. Box & Whiskers plot displaying mean and median of data collected

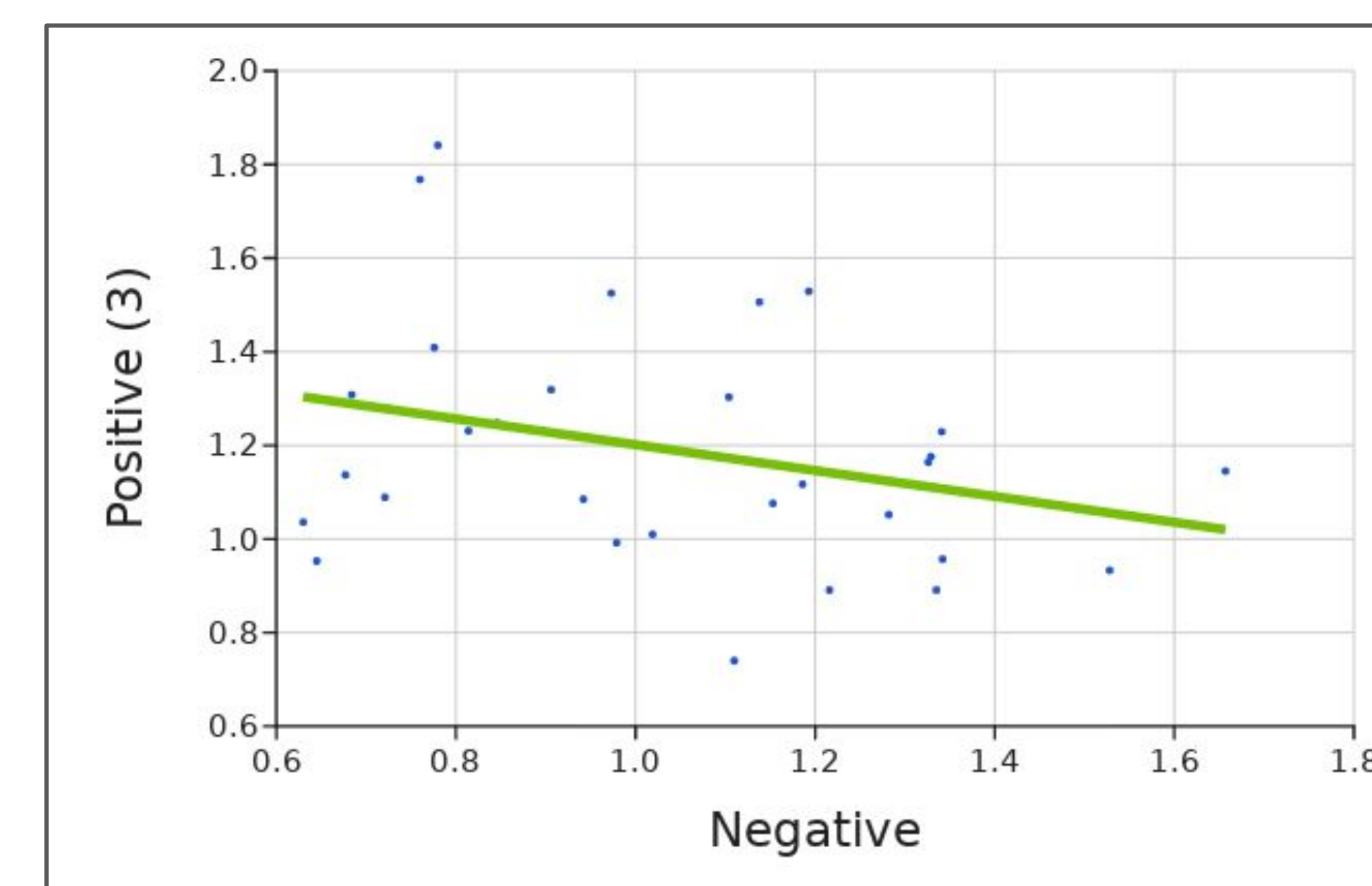


Chart created by finalists using DataClassroom

Fig 5. Relationship comparison graph between Negative and Positive 1 with a line of regression

References

- Azeem, I., Adeel, M., Ahmad, M. A., Shakoor, N., Jiangcuo, G. D., Azeem, K., Ishfaq, A., Shakoor, A., Ayaz, M., Xu, M., & Rui, Y. (2021, November 1). Uptake and accumulation of nano/microplastics in plants: A critical review. *Nanomaterials*, 11(11), 2935. <https://doi.org/10.3390/nano11112935>
- Jia, L., Liu, L., Zhang, Y., Fu, W., Liu, X., Wang, Q., Tanveer, M., & Huang, L. (2023, August 10). Microplastic stress in plants: effects on plant growth and their remediations. *Frontiers in Plant Science*, 14, 1226484. <https://doi.org/10.3389/fpls.2023.1226484>
- Kadac-Czapska, K., Ośko, J., Knez, E., & Grembecka, M. (2024, May 7). Microplastics and oxidative stress-current problems and prospects. *Antioxidants*, 13(5), 579. <https://doi.org/10.3390/antiox13050579>